

EMERGENT CAUSALITIES - FMSM - Emergent Causalities Framework

A Proposal for a Structured Protocol for Language-Model-Assisted Multidomain Analysis

Abstract

Economic, organizational, technological, social, and political events rarely result from a single cause. They generally arise from interactions among multiple mechanisms, actors, constraints, and decisions whose effects may reinforce one another, offset one another, or emerge only after several stages.

Language models can rapidly draw on knowledge from different domains. This capability also presents a challenge: a coherent explanation does not guarantee the validity of either the information used or the proposed causal relationships. A model may construct a convincing narrative from a false premise, settle prematurely on one explanation, attribute an undocumented intention to an actor, or gradually turn a hypothesis into near-certainty.

EMERGENT CAUSALITIES - Emergent Causalities Framework is an experimental proposal for structuring this type of analysis. It is not intended to serve as a truth engine, a prediction system, or an automated substitute for domain expertise. It seeks to impose an identifiable analytical procedure that separates observations from interpretations, identifies mechanisms that may connect multiple phenomena, preserves competing hypotheses, and makes uncertainty and counter-evidence visible.

Its analytical core is a four-level decomposition - **Facts, Mechanisms, Strategies, and Motivations (FMSM)** - supplemented by several control operations: premise qualification and neutralization, epistemic-status preservation, distance from the premise, exploration of competing hypotheses, and the search for counter-evidence.

This paper presents the framework's principles, explains how they fit together, and reports the initial behaviors observed in a series of exploratory tests conducted on different language models.

1. Why an Emergent Causalities Framework?

An analysis often begins with a visible event:

- a company loses customers;
- an organization experiences rising employee departures;
- a technology is rapidly adopted;
- a decision produces an unexpected outcome.

The most readily available explanation is then to look for a cause directly associated with the phenomenon.

This representation may be sufficient for some simple problems. It becomes fragile, however, when several domains interact.

A decline in store traffic may depend simultaneously on price, accessibility, competition, consumer habits, or seasonal factors. A productivity gain observed after remote work is introduced may likewise coincide with new tools, a change in management, or a major shift in the market.

The problem, therefore, is not merely a lack of information. It is the need to **organize relationships among available information without prematurely assigning a causal role that one element may not actually have.**

Language models make this difficulty especially visible. Their ability to produce coherent explanations is both a strength and a source of analytical risk: a narrative may be exceptionally well constructed while resting on incorrect information or an insufficiently established relationship.

Emergent Causalities therefore begins with a simple proposition:

Multidomain analysis benefits from explicitly distinguishing what is observed, the processes proposed to explain it, the actors' actions, and the motivations that may underlie those actions.

The aim is not to eliminate uncertainty, but to give it an identifiable place within the analysis.

2. General Architecture

Emergent Causalities treats analysis as a process of **divergence followed by controlled convergence.**

In the first phase, the system explores multiple elements that may contribute to the phenomenon: actors, constraints, relevant domains, precedents, possible mechanisms, consequences, and competing explanations.

This exploration is not a conclusion.

An element generated during this phase does not become more probable merely because it fits elegantly into the rest of the reasoning. The system must then qualify, compare, and, where appropriate, eliminate the proposals it has produced.

The analysis gradually converges on a structured representation that includes:

- the available facts and observations;
- the proposed dominant mechanisms;
- the observed or possible strategies;
- the system's leverage points and vulnerabilities;
- the mismatches and points of friction;
- the competing hypotheses;

- the coherence of the proposed trajectories;
- the uncertainties;
- the counter-evidence.

This representation is a **form of output**. It should not be confused with the full set of operations required to produce it. This distinction allows the framework's internal operation to evolve without imposing the same degree of complexity on its users.

3. FMSM: Four Levels That Must Remain Distinct

FMSM provides the central decomposition framework.

3.1 Facts - What Is Observed or Given?

The first level includes what is observed, documented, supplied by the user, or explicitly imposed as a condition.

This final distinction is important.

The statement:

"Danny DeVito plays the Terminator in 1984"

is a proposition about historical reality and can be evaluated as such.

By contrast:

"What if Danny DeVito had played the Terminator in 1984?"

establishes a counterfactual condition.

It is not converted into a historical fact. It does, however, become a valid premise **within the requested hypothetical space**.

The first level therefore does more than collect information: it identifies **the nature of what is given to the system**.

3.2 Mechanisms - Through What Process Can A Affect B?

Mechanisms are explanatory processes that may recur across multiple domains.

Transmission, selection, cooperation, competition, feedback, adaptation, learning, optimization, and control are examples of process families that can be observed in very different environments.

One distinction is fundamental:

A mechanism is not an intention.

A network effect may emerge without any actor deciding to create it. A feedback loop may form without being understood by the individuals participating in it.

Emergent Causalities therefore seeks to explain certain system behaviors before assuming why an actor might have wanted to produce them.

3.3 Strategies - What Does the Actor Do in Response to the System?

A strategy is an action or set of actions through which an actor attempts to intervene in the identified mechanisms and constraints.

The actor may try to exploit a mechanism, circumvent it, adapt to it, neutralize it, or trigger another mechanism.

This separation makes it possible to distinguish **the justification for an action from its effectiveness**.

A defensible motivation may lead to an ineffective or catastrophic strategy. Conversely, a strategy's effectiveness does not make its underlying motivation acceptable.

Judging the actor and evaluating the strategy thus become two distinct analytical problems.

3.4 Motivations - Why Might the Actor Act This Way?

The fourth level is deliberately the most tentative.

Security, status, economic interest, belonging, power, preservation of a value, or organizational survival may help explain a behavior. Observable behavior alone, however, cannot establish a motivation with certainty.

Emergent Causalities therefore favors:

observable behavior -> several compatible motivations

rather than:

observable behavior -> certain motivation.

This distinction also separates **internal coherence from external validity**. An actor may consider an action perfectly rational because it is consistent with their representation of the situation, even when that representation is incorrect.

4. Premise Qualification and Neutralization

Before explaining a phenomenon, Emergent Causalities seeks to determine what the question actually contains.

A seemingly simple formulation may combine:

an event;

an interpretation;

an attributed intention;

an assumed causal link;

a prediction;
a hypothetical condition.

Consider:

"Sony is abandoning physical media to control its customers."

This sentence contains at least several distinct propositions:

Event: reduction or abandonment of physical media.

Possible consequence: greater control over distribution.

Attributed motivation: desire for control.

Proposed causal link: dematerialization -> greater control.

These elements do not automatically share the same status.

Qualification identifies their nature.

Neutralization performs a different operation: it removes wording that could artificially constrain the conclusion.

The neutralized question then becomes: **"Why is Sony abandoning physical media?"**

Another example:

"Why was this catastrophic decision made?"

should not lead the system to treat *catastrophic* as an already established property of the decision.

The decision and its consequences must first be examined.

5. Principle of Epistemic-Status Preservation

Each piece of information has a status, which may include:

established/documented; supplied as a premise; inferred; hypothetical; uncertain; disputed.

A common error in complex reasoning is to let this distinction gradually disappear.

A hypothesis is introduced. Several paragraphs are built around it. Its coherence with other elements increases. A few steps later, it is restated as though it had been demonstrated.

For this reason, Emergent Causalities introduces a **principle of epistemic-status preservation**:

Information cannot spontaneously acquire a higher epistemic status during the analysis.

A hypothesis may be strengthened or weakened. It does not, however, become a fact merely because a coherent narrative has been constructed around it. Promotion to a higher status requires new information capable of supporting that change.

This principle does not guarantee the accuracy of the content being processed. It seeks to prevent **the reasoning process itself from artificially increasing the degree of certainty assigned to its premises.**

6. Distance from the Premise

Not all proposed consequences lie at the same distance from the initial observation.

Emergent Causalities uses a simple representation:

D0 - Initial observation or premise

D1 - Relatively direct consequence, requiring few intermediate steps.

D2 - Intermediate consequence, requiring one or more additional mechanisms.

D3+ - Distant systemic propagation, dependent on a larger number of conditions.

This classification does not mean:

D3 = false or improbable.

It means:

As distance increases, the trajectory's dependencies and associated uncertainty must become more visible.

A distant trajectory may well occur. It simply requires more conditions to reach its final state.

This representation also makes it possible to treat foresight not as categorical prediction, but as a conditional structure:

If A remains true, if M continues to operate, and if C does not intervene, then B is a compatible trajectory.

7. Competing Hypotheses and Resistance to Premature Closure

A convincing explanation can become a trap.

As soon as one mechanism appears to explain several observations, it becomes tempting to organize all subsequent information around it.

Emergent Causalities instead seeks to preserve several compatible explanations temporarily.

For each hypothesis, the analysis may look for:

supporting evidence; disconfirming evidence; necessary conditions; information that could distinguish among the hypotheses.

The coexistence of several hypotheses does not constitute additional evidence for any one of them.

Two other categories then come into play.

Uncertainty

Necessary information that is currently unavailable.

Counter-evidence

Available information that weakens a hypothesis.

The two are not interchangeable. Not knowing an organization's internal costs is an uncertainty. Observing behavior that is incompatible with a proposed explanation may constitute counter-evidence.

The search for counter-evidence introduces a question that complements the conventional search for confirmation:

What should we observe if our explanation were false?

This operation is intended in particular to limit the construction of an excessively coherent narrative around the first satisfactory explanation.

8. Leverage Points, Vulnerabilities, and Points of Friction

Once the main mechanisms have been identified, Emergent Causalities distinguishes several properties of the system.

A **mechanism** is a process.

A **leverage point** is a point of intervention whose modification can produce a significant effect.

A **strategy** is the way an actor uses or combines leverage points and mechanisms to pursue an objective.

A **vulnerability** is an element whose modification or failure may impair a trajectory. It need not be failing at present: a critical dependency that functions perfectly is still a vulnerability.

Finally, **mismatches or points of friction** are places where several elements of the system do not fit together as expected:

discourse <-> behavior;

objective <-> outcome;

strategy <-> environment;

model <-> conditions on the ground;

short term <-> long term.

These mismatches may indicate a missing mechanism, a flawed hypothesis, or a transformation of the system that has not yet been represented correctly.

9. Feedback Loops and Trajectories

Some causal relationships do not form a linear chain.

A changes B, and B then acts back on A.

Feedback may amplify a dynamic or help stabilize it. The term *positive*, when applied to an amplifying loop, therefore does not mean that the loop is desirable.

These structures help explain how an initially limited intervention can gradually alter the system that produced it.

Emergent Causalities also uses the concept of **trajectory coherence**.

It does not answer:

"Will this happen?"

but rather:

"Can the identified observations, mechanisms, and strategies form part of the same development without major contradiction?"

This yields an important distinction:

coherence != probability.

A trajectory may be highly coherent while remaining very improbable.

10. Special Case: Counterfactual Analysis

The initial PoC tests revealed a particular difficulty.

When the model was given a premise that clearly conflicted with its knowledge, the verification mechanism could become so dominant that the system effectively refused to explore the requested scenario.

This observation led to a distinction among three situations:

factual statement -> its real-world status must be preserved;

plausible hypothesis -> it may be analyzed conditionally;

explicitly imposed counterfactual -> the premise may be accepted locally without being converted into a fact about the real world.

The resulting principle can be stated as follows:

Suspend verification of the premise within the scenario without altering its status outside the scenario.

A scenario in which Danny DeVito replaces Arnold Schwarzenegger in *The Terminator* can thus be analyzed as a counterfactual without causing the system to claim that the event actually occurred.

This function is also useful for prospective scenarios, decision simulations, and the exploration of deliberately improbable events.

11. Initial Exploratory Tests of the PoC

The PoC was applied to several categories of problems to observe its behavior rather than presume its effectiveness.

These trials do not constitute formal experimental validation. They nevertheless help identify behaviors, limitations, and failure modes that can inform the framework's development.

Test	Primary challenge	Target behavior
Single supplier	Causal propagation	Identify dependency, buffer, and propagation
Restaurant	Multifactoriality	Preserve several compatible causes
Training	Feedback	Identify a degradation loop
Ice cream / drownings	Correlation	Look for a confounding variable
Remote work	Causal attribution	Reject a single explanation without controls
Departures / hiring	Temporality	Detect a dynamic that predates the alleged cause
DeVito / Terminator	Counterfactual	Locally accept a premise that is false in the real world
Musk / Lada	Improbable scenario	Construct conditional consequences
AI laboratory	Complex foresight	Produce several organizational trajectories

These tests showed, among other things, that the imposed structure can lead the model to preserve several explanations simultaneously and specify the data required to distinguish among them.

They also showed that the quality of the output depends heavily on the model used.

This observation is an important limitation, not a result to be concealed.

12. Influence of the Underlying Model

The exploratory trials were conducted on several language models with different sizes and capabilities. The results show that applying the same protocol does not produce the same analytical quality across models.

Models with more limited capabilities may follow the requested structure correctly while introducing more factual errors, approximations, or weak causal relationships. More capable models, by contrast, may use the same structure to produce richer competing hypotheses, identify more dependencies, and preserve epistemic distinctions more effectively.

This observation suggests separating two dimensions:

Model capability - the quality and breadth of its knowledge, its reasoning, its ability to maintain context, and its capacity to handle several relationships simultaneously.

Emergent Causalities protocol - the organization imposed on the processing of those capabilities.

The PoC does not currently allow the contribution of each dimension to be quantified independently. A controlled comparison among models, using identical test sets and predefined evaluation criteria, would be a subsequent validation step.

13. Observed Failure Modes

Trials conducted with different models revealed several categories of possible failure.

Factual hallucination

The model uses incorrect information as its starting point.

Semantic drift

The model initially addresses the requested problem, then drifts toward an adjacent topic.

Rule overactivation

A useful mechanism or instruction becomes dominant and is applied in situations where it contributes little or degrades the analysis.

Epistemic overcorrection

The system becomes so cautious that it refuses an exercise that is explicitly hypothetical.

Excessive propagation

A hypothesis several steps removed from the initial observation gradually acquires excessive importance and ultimately structures the conclusion.

These behaviors suggest a fundamental distinction:

The quality of the analytical procedure and the quality of the model's knowledge are two different variables.

A protocol may improve the organization of reasoning without correcting an inaccurate knowledge base.

14. What Emergent Causalities Does Not Demonstrate

The scope of the PoC must be explicitly delimited.

Emergent Causalities is not a truth engine

A highly structured analysis may still be wrong if its input data are incorrect.

Emergent Causalities seeks to improve the organization of reasoning, not to guarantee the truth of its premises or conclusions.

Emergent Causalities is not a predictor

It produces conditional trajectories, not certainties about the future.

Emergent Causalities does not replace domain expertise

Connecting economics, psychology, technology, or sociology does not turn the system into an economist, psychologist, engineer, or sociologist.

Its function is different: to find and organize possible relationships across these domains.

Emergent Causalities does not guarantee impartiality

Neutralization, competing hypotheses, and counter-evidence are operations intended to limit certain forms of premature closure or unjustified attribution.

A biased model can nevertheless produce a biased analysis under Emergent Causalities.

Emergent Causalities does not guarantee textual reproducibility

Because language models are stochastic and different models have different capabilities, the same input will not necessarily produce the same text.

The reasonable objective is narrower: **to make an analytical procedure identifiable and allow comparison among similar control structures.**

15. Discussion

The initial Emergent Causalities trials suggest that its potential value lies not in adding a new source of knowledge to the model, but in organizing the operations it performs on the knowledge already available to it.

This distinction is important.

A model with more knowledge or reasoning capacity may produce a richer analysis. Emergent Causalities does not seek to eliminate this difference among models. Instead, it provides a common set of analytical constraints: separate information by status, make proposed relationships explicit, preserve uncertainty, and prevent the implicit promotion of hypotheses.

The trials also show that a useful constraint may become harmful when applied too rigidly. The initial handling of counterfactual scenarios provides one example: premise verification, appropriate for a factual statement, prevented the exploration of a deliberately unreal scenario.

The framework must therefore balance two symmetrical errors:

accepting a premise too readily and building on false information;

and

rejecting uncertainty so strongly that no hypothetical exploration remains possible.

This tension is likely to be one of the central problems for future versions to examine.

16. Research and Development Directions

Future work could include:

- Developing a reproducible evaluation protocol for comparisons among models;
- Measuring the framework's independent effect by comparing models used with and without the protocol;
- Systematically studying failure modes;
- Evaluating the stability of analyses across repeated runs;
- Extending tests to different domains and degrees of causal complexity;
- Improving the representation of dependencies, uncertainties, and long trajectories.

17. Conclusion

Emergent Causalities arose from an observation: apparently distant domains may exhibit comparable mechanisms, while the same visible event may result from several simultaneous causal chains. The framework was subsequently developed as a demonstration tool for the book of the same name.

Its experimental ambition is to give a language model a procedure for navigating this space without systematically confusing observation, explanation, strategy, and motivation.

FMSM provides one initial axis for this separation. Premise qualification, the principle of epistemic-status preservation, distance from the premise, competing hypotheses, counter-

evidence, and feedback analysis are complementary mechanisms intended to preserve the structure of the reasoning as its complexity increases.

The initial PoC tests reveal both its value and its limitations. It can produce structured analyses of simple, multifactorial, counterfactual, or prospective causal problems. Its quality, however, remains dependent on the knowledge, capabilities, and biases of the model executing it.

This dependence is not external to the problem; it is part of the problem.

Emergent Causalities therefore does not propose turning a language model into an arbiter of reality. It proposes something more limited:

Making more visible the steps through which an observation becomes an explanation and then a possible trajectory.

The value of such a representation does not necessarily lie in the certainty of its conclusion. It also lies in the ability to see where that conclusion depends on a hypothesis, which mechanism is assumed to connect two events, what information might challenge it, and where our knowledge of the system ends.

In a complex environment, recognizing this boundary is not a failure of analysis; it is part of the analysis itself.